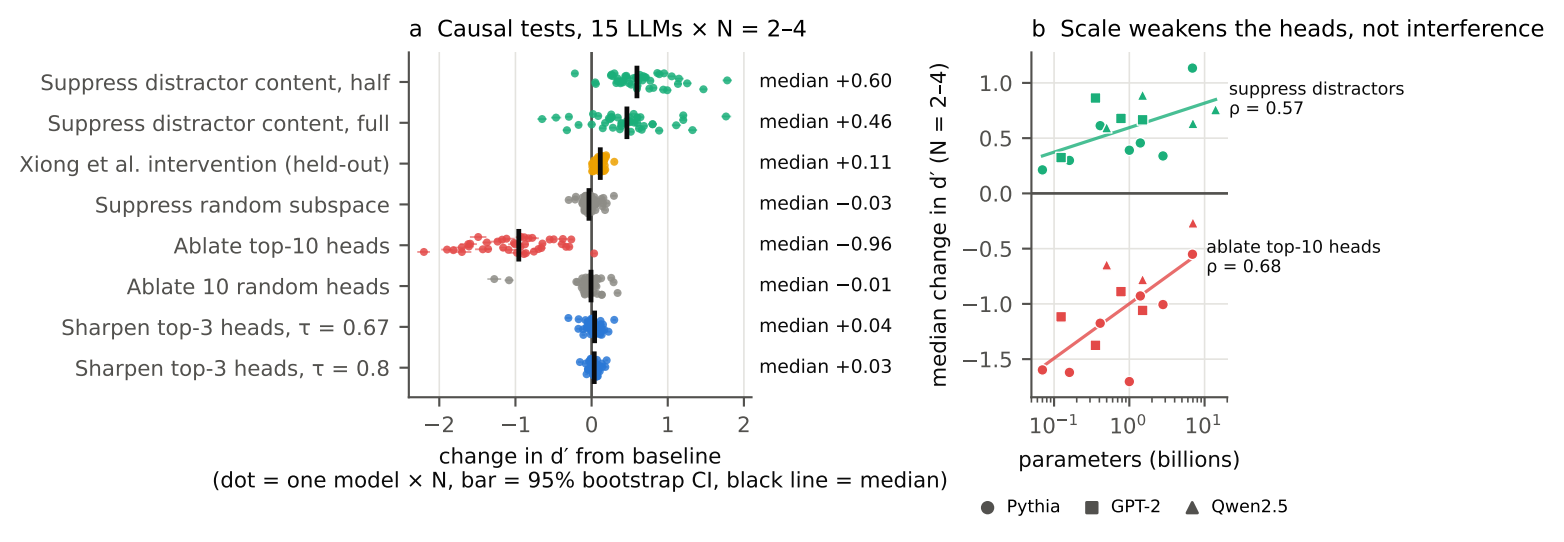
a Causal tests, 15 LLMs × N = 2–4
Suppress distractor content, half
Suppress distractor content, full
Xiong et al. intervention (held-out)
Suppress random subspace
Ablate top-10 heads
Ablate 10 random heads
Sharpen top-3 heads, τ = 0.67
Sharpen top-3 heads, τ = 0.8
median +0.60
median +0.46
median +0.11
median −0.03
median −0.96
median −0.01
median +0.04
median +0.03
−2
−1
0
1
2
change in d′ from baseline
(dot = one model × N, bar = 95% bootstrap CI, black line = median)
b Scale weakens the heads, not interference
median change in d′ (N = 2–4)
1.0
0.5
0.0
−0.5
−1.0
−1.5
suppress distractors
ρ = 0.57
ablate top-10 heads
ρ = 0.68
10^{-1}
10^{0}
10^{1}
parameters (billions)
Pythia
GPT-2
Qwen2.5